5. Induction

The one idea the whole machine rests on

A *changing* magnetic field makes a voltage

Lesson 5 · about 90 minutes · everyone builds; no external power at all

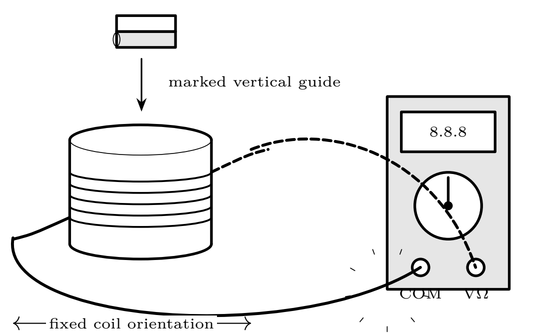
What you need

- Magnet wire, 26–30 AWG, 30 m or more
- A cardboard tube (kitchen roll) and a length of copper pipe
- Strong neodymium magnets that drop freely through both
- Multimeter with a millivolt range
- Two LEDs wired back-to-back
- An aluminium tube, if you have one

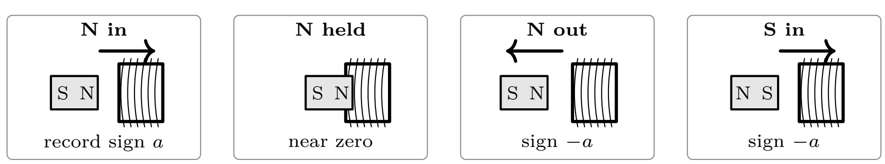
Predict first

You drop a magnet down a copper pipe. Copper is not magnetic and the magnet never touches the walls. Does it fall faster than, slower than, or the same as a magnet dropped down a cardboard tube of the same size? Commit before you drop anything.

How to set it up



200–400 turns; meter leads, coil face, start height, and travel line stay fixed



reverse motion or pole: reverse sign; reverse both: preserve sign
The letter a avoids guessing +: your fixed lead convention determines its name.

Do it

1. **Make the apparatus repeatable.** Wind 200–400 tidy turns, secure them, sand only the two ends, and connect to the voltage jacks on the lowest suitable DC millivolt range. Mark one coil face, the magnet's N end, and start–center–finish positions. Ask: *what must not change if signs are to be comparable?*
2. **Establish the quiet reading.** With the magnet far away, watch for ten seconds and record the largest drift from zero. Wiggle each lead without moving the magnet. If that makes a pulse, repair the connection before proceeding. This drift is the experiment's practical zero band.

3. **Separate field from change.** Hold N still at start, center, and finish for three seconds each. Record the settled reading. Ask at every stop: *the field differs here, but is the linked flux changing now?*
4. **Measure N entering.** Move start to finish on a slow three-count. Record peak magnitude and sign. Return the magnet well outside the coil without recording. Repeat five times along the same line. Report the median, minimum, maximum, and half-range. Ask: *is the spread smaller than the difference you hope to claim later?*
5. **Reverse motion.** Pull N from finish to start on the same count, five times. Predict before every pass. Compare signs first, then magnitudes. Ask: *did reversing motion reverse the change in linked flux? Did each stopped state return inside the zero band?*
6. **Change rate, not route.** Repeat N-in and N-out on a quick one-count, five times each. Keep endpoints, pole, coil, and meter unchanged. Compare median peaks with the slow sets. Do not call a small difference real if the ranges overlap heavily.
7. **Reverse the field.** Turn the magnet end-for-end without swapping leads. Collect five S-in and five S-out trials at the slow count. Ask: *which sign matches N-out? What must happen when both pole and motion reverse?*
8. **Final proof.** In the order N-in, N-out, S-in, S-out, perform two slow passes of each. Before moving, write the predicted sign using the first N-in sign as a. The proof passes only if each pair repeats, every stop returns inside the zero band, reversing either pole or motion reverses sign, and reversing both preserves it.
9. **Troubleshoot before interpreting.** An erratic or absent result means: check wrong meter jack or range, unsanded enamel, loose clips, broken winding, auto-ranging delay, changed coil face, ambiguous magnet pole, sideways motion, different endpoints or speed, nearby steel, and moving leads. Change one suspected cause, re-establish the zero band, then repeat the entire proof—never select only the favorable trials.
10. **Make the effects visible.** Replace the meter with two LEDs back-to-back. Move briskly; one may flash on entry and the other on exit. Failure to see a flash does not overturn the meter data: LED threshold, ambient light, coil resistance, and speed matter.
11. **Test magnetic braking.** Time at least five drops through matched cardboard and copper pipes from one marked release height; use median and range. The magnet must fall freely without scraping. Repeat with aluminium if available. Ask: *how can a nonmagnetic conductor oppose motion without touching the magnet?*

What it means

Faraday's law	A voltage appears in a loop when the magnetic field *through* that loop changes. Not when it is strong — when it *changes*. Faster change, bigger voltage; more turns, bigger voltage.
Why holding it still gives nothing	This is the whole lesson. A stationary magnet inside a coil produces no sustained induced voltage; a real meter may retain a small offset or drift inside the measured zero band. Every generator, every transformer and every Tesla coil exists to make something change.
Sign is relational	A meter's + sign depends on which coil end reaches its red lead. Fix that convention, then the physical rule is robust: reverse motion or pole and sign reverses; reverse both and sign is preserved.
Lenz's law	The opposite signs represent an induced effect that opposes the change that made it. Push a magnet toward a closed coil and the coil's field resists the approach; pull away and it resists the departure.
The copper pipe	The falling magnet induces circulating eddy currents in the pipe wall. Those currents make their own magnetic field, which by Lenz's law opposes the fall. The magnet falls in slow motion. Nothing touches it and copper is not magnetic — it is being braked by a field that its own falling created. This is the most surprising thing in the course, and it is worth doing several times.
Conservation	The energy in the flash came from your arm. Moving a magnet into a coil is *harder* when the circuit is closed than when it is open. Try it with the LEDs connected and disconnected and feel the difference.

Connection forward. A Tesla coil places its many-turn secondary in a rapidly changing primary field; tuning lets successive cycles build. Lessons 6, 8 and 9 add turns ratio, oscillation, and coupling, but every clause still depends on *change*, not field strength alone.

> **This lesson is low voltage.** Everything here runs from batteries or a small plug-in supply and cannot hurt you. That changes at Lesson 10. Build the habits now — one hand, discharge before touching, check before you power up — while the stakes are nothing, so that they are automatic when the stakes are real.